Kaustav Tarafdar

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Backend & AI Infrastructure Engineer - 2 years building distributed systems, ML inference infrastructure and multi-tenant SaaS platforms in Python and TypeScript. Scaled GPU inference to 50 requests/sec across 10 nodes, cut inference latency 200x, and owned the backend for a consumer app launch to 80K users as the second engineer on a founding team.

Work Experience

Software Engineer, TechyMonks techymonks.com Feb 2026 – Present

- **Led the v2 re-architecture** of a wholesale deal-evaluation platform (Node.js, TypeScript, PostgreSQL), migrating to a Controller-Service-Repository pattern, normalizing a fragmented schema and eliminating N+1 queries.
- **Launched the platform's affiliate revenue channel** end-to-end, automating commission tracking through Stripe webhooks and replacing manual payout handling with saved banking details, autopay and bulk admin approvals.
- **Architected an HVAC lead-conversion backend** so tenants could keep running on their existing CRM, integrating 3 providers (ServiceTitan, CallTrackingMetrics, GoHighLevel) over a polling and webhook pipeline with PostgreSQL schema-per-tenant isolation.
- **Owned the mobile sales-app backend** serving field reps their daily appointments, AI call summaries and geo-fenced arrival alerts, plus Twilio masked calling and an LLM agent for company analytics.
- **Replaced manual salon bookings** with a multi-tenant FastAPI platform driving a webhook-based WhatsApp bot over Redis, PostgreSQL and S3, with LangChain campaign generation and customer-insight engines on Gemini and OpenAI.
- **Built CI/CD from scratch** with GitHub Actions, replacing manual releases with isolated dev/stage/prod pipelines sequencing build, migrate and deploy behind ESLint/TypeScript gates and New Relic/Papertrail observability.

Software Engineer, Founding Team, Yonder Wonder AI yonderwonder.ai Mar 2025 – Feb 2026

- **Architected a distributed GPU inference platform** across 10 GPU VMs (160GB VRAM) using FastAPI, Redis RQ and Docker, supporting 40 concurrent video runs at peak load of 50 requests/sec; cut idle GPU spend by reaping orphaned processes on client disconnect.
- **Cut per-image inference from 2s to ~0.01s (200x)** by profiling model timings and removing redundant image-to-byte conversions; extended codec support to 7 formats via ffmpeg and H.264 handling.
- **Replaced a Flask monolith** with an async FastAPI queue and ML-worker architecture on Redis RQ with dead-letter queues, and led the GCP to Azure migration including Azure Blob Storage.
- **Diagnosed a production memory leak** with GNU Debugger - a PyTorch pointer deallocation across shared threads - eliminating runtime crashes as the platform scaled to 5,000 users.
- **Sole backend owner of WeClick**: authentication, in-app credits with iOS/Android payment SDKs, and autoscaling inference workers on Azure Container Apps for an 80K-user launch; cut deploys from 10 min to 2 min.

Backend Engineer Intern, Agetware agetware.com Nov 2024 – Feb 2025

- **Built a job application management system end-to-end** for a music-school client: a state machine covering submission, interview scheduling and hire, with AWS S3 uploads for resumes, videos and photos.
- **Integrated Google Calendar scheduling and WhatsApp notifications** for applicants and interviewers, replacing a manual hiring workflow.
- **Streamlined class scheduling** in the admin dashboard, replacing a multi-step Google Calendar workflow with one-click student-to-class assignment.

Projects

API Gateway GitHub

- Built a self-hosted API gateway with service discovery, consistent-hash load balancing, health-aware failover and dynamic routing; added authentication, structured logging, Docker deployment and 30+ unit and integration tests.

URL Shortener GitHub

- Built a URL shortening platform supporting custom aliases, QR generation and click analytics with frequency and recency tracking in PostgreSQL.

Education

B.Tech in Computer Science & Engineering, MAKAUT (FIEM), Kolkata 2021–2025

Technical Skills

- **Languages:** Python, TypeScript, JavaScript, SQL, Bash
- **Backend:** FastAPI, Django, Flask, Node.js, Express.js, GraphQL, REST APIs, WebSockets
- **Databases:** PostgreSQL, Redis, MongoDB, Supabase, Database Design & Normalization
- **AI/ML:** LangChain, OpenAI, Gemini, PyTorch, GPU Orchestration, Model Inference Pipelines
- **Cloud:** AWS (S3, EC2), Azure (Container Apps, Blob Storage), GCP, Heroku, Fly.io
- **DevOps:** Docker, GitHub Actions, CI/CD, Git, Nginx, Linux, Monitoring & Observability (New Relic, Papertrail), Unit & Integration Testing
- **Architecture:** Distributed Systems, System Design, API Design, Microservices, Event-Driven Architecture, Message Queues (Redis RQ, BullMQ, Celery), Multi-Tenancy, Domain-Driven Design, Scalability, Performance Optimization
- **Integrations:** Stripe, Twilio, WhatsApp/Meta APIs, REST, Webhooks, OAuth
- **Practices:** Technical Documentation, Code Review, Cross-Functional Collaboration